

Generation Interconnection System Impact Study Report

For Queue Project Q24-2395

Arcogen 138 kV
January 20, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2395. The Interconnection Customer proposes a 56.5 MW battery energy storage facility interconnecting at the Arcogen 138 kV substation in the Virginia area. The request was received on March 25, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 546 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$882,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ARCOGEN – CHINCOTEAGUE 138 kV line	TASLEY – SOUTHTOWN 345 kV	119.0	386	89.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHINCOTEAGUE – ARCOGEN 138 kV	Rebuild line with higher-capacity conductor	\$8,442,000	10% (\$882,000)
	Total allocated to project		\$882,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.